

Does It Rain More in Seattle Than Berkeley?

Introduction

Seattle is widely known as a rainy city, but how much wetter is it than another West Coast city? This project compares precipitation in Seattle, Washington, and Berkeley, California. The two cities are located at similar longitudes, but Seattle is much farther north and has a different climate. The main question of this analysis is: **Does it rain more in Seattle than in Berkeley?**

Data and Methods

Daily precipitation data were obtained from the NOAA National Centers for Environmental Information using the Global Historical Climatology Network Daily dataset. The main analysis covers January 1, 2018 through December 31, 2022.

The Seattle and Berkeley datasets were cleaned and combined so that both cities had one precipitation value for every day in the five-year period. Berkeley had 471 missing precipitation values, while Seattle had 190.

Because Berkeley had several long periods of missing data, historical Berkeley weather data were used to estimate those missing values. A method based on precipitation from similar times of the year was chosen because it better preserved Berkeley's strong seasonal pattern during long gaps. Seattle's fewer missing values were estimated using the average precipitation for the same day of the year across the available years.

After cleaning the data, I compared overall daily precipitation, monthly patterns, and seasonal precipitation. Statistical tests were also used to determine whether the differences between the cities were large enough to be considered meaningful rather than simply random variation.

Results

Seattle had substantially higher average daily precipitation over the five-year period. Seattle averaged approximately 0.113 inches per day, while Berkeley averaged approximately 0.053 inches per day.

The monthly comparison shows that the difference is especially noticeable during fall and winter. Berkeley becomes extremely dry during the summer, while Seattle continues to receive some precipitation.

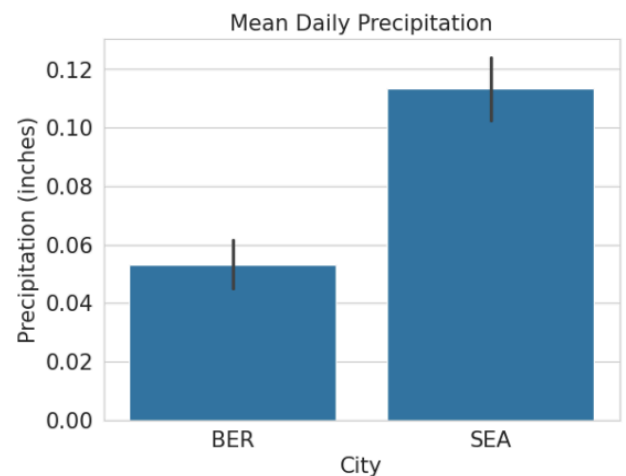


Figure 1. Mean Daily Precipitation: Seattle vs. Berkeley

Statistical tests of monthly precipitation found significant differences between Seattle and Berkeley in 10 of the 12 months. March and April were the only months where the analysis did not find a statistically significant difference at the 0.05 level.

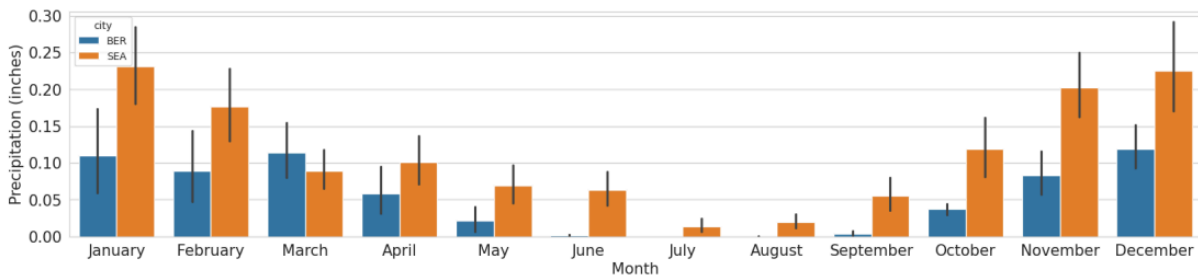


Figure 2. Mean Monthly Precipitation by City

The seasonal analysis showed a similar pattern. Seattle averaged 19.10 inches in winter compared with 9.62 inches in Berkeley, and 11.40 inches in fall compared with 3.77 inches in Berkeley. The largest relative difference occurred in summer: Seattle averaged 2.95 inches, while Berkeley averaged only 0.06 inches.

The seasonal differences were statistically significant in winter, summer, and fall. Spring was the only season where the difference was not statistically significant.

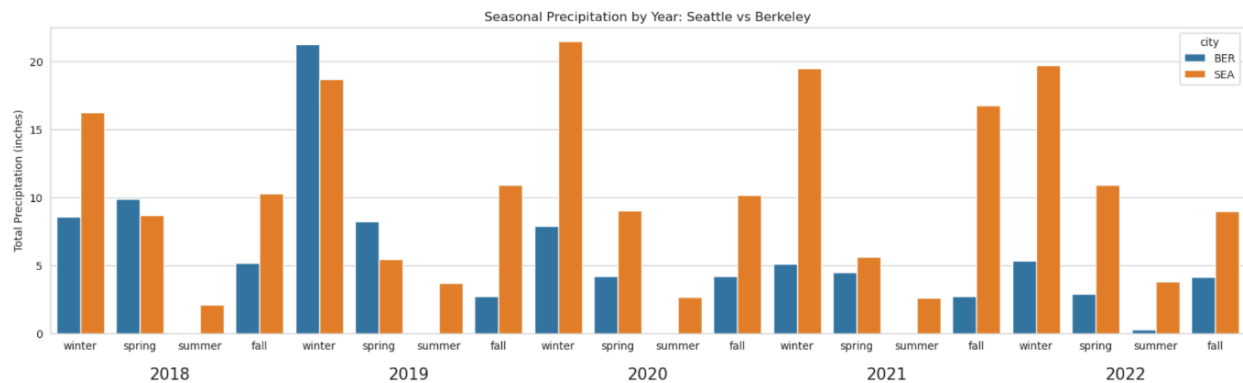


Figure 3. Seasonal Precipitation by Year: Seattle vs. Berkeley

Conclusion

The results support the conclusion that it rains more in Seattle than in Berkeley. Seattle had more than twice the average daily precipitation over the five-year study period. The difference was also visible throughout most months and seasons.

The contrast was strongest during summer, when Berkeley received almost no precipitation while Seattle continued to receive some rain. Although the two cities occasionally experienced similar conditions, the overall daily, monthly, and seasonal results consistently show that Seattle is the wetter city.